

From Endpoint Sufficiency to Structural Gaps: Clarifying the Framework-Level Content of a Cross-Level Diagnostic Program

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Abstract

This article clarifies the framework-level content of a cross-level diagnostic structure developed across three companion papers on endpoint-path gaps, declared structural residuals, and fixed-slice fiber diagnostics. Its aim is not to re-prove those papers one by one, but to state the shared structural problem they address and the diagnostic framework they jointly establish.

The recurring difficulty is straightforward. A coarse description is often treated as sufficient before one has declared what is visible, admissible, comparable, reconstructible, or negligible under the report. That omission is structural rather than merely technical. Sufficiency, residual ambiguity, and approximation are therefore always conditional on a locked viewpoint/package.

Under that condition, the three papers occupy distinct slices of one framework. Paper I supplies the strict finite-window anchor for endpoint sufficiency and endpoint-path residuals. Paper II extracts the general diagnostic framework of viewpoint/package locking, the distinction between within-level tension and cross-level residual structure, and a minimal static-to-dynamic interface. Paper III shows that the same pressure reappears in a fixed-slice setting, where histories sharing an endpoint may still exhibit predictive spread inside endpoint fibers.

The contribution of this overview is therefore not merely synthetic. It makes explicit what is already stable across the three papers at framework level: viewpoint/package locking as a condition of determinate cross-level reporting; Δ as an explicit carrier of non-closure rather than a generic error term; the ψ - Δ separation as a distinction between within-level tension and cross-level residual structure; and an output space that includes near-zero, non-negligible, package-locked, and non-licensed outcomes. It also clarifies why the declared viewpoint/package has philosophical weight: V is not an external annotation added to an otherwise complete report, but the coupling layer under which structure becomes reportable at all. The article should therefore be read as a clarification of what the companion papers already formalize at framework level. It does not divide domains into those already unified and those waiting to be unified later. The same cross-level structural pattern appears in logic, physics, computation, and learning. What varies from case to case is the local carrier, the declared package, the admissible reconstruction discipline, the reporting resolution, and whether the problem is handled by direct structural diagnosis or by a further computable realization.

Keywords: cross-level diagnostics; endpoint sufficiency; structural gaps; declared viewpoints; residual structure; hierarchical attention

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1 Introduction: Why a Cross-Level Diagnostic Framework Became Necessary

1.1 Recurrent Difficulties in Cross-Level Reasoning

Across logic, physics, computation, and learning, the same difficulty recurs. A coarse description works for a task and is then over-read as if it exhausted the relevant structure. The object may be an endpoint, a macroscopic label, a compressed state, or a protocol-visible output. Sometimes that compression is licensed. Sometimes a residual remains active under the declared task even though it no longer appears at the coarse level.

The problem is not coarse description itself, but the omission of the conditions under which the report is made. Cross-level judgments are often issued before one has fixed what is visible, admissible, comparable, reconstructible, or negligible. Once those clauses remain implicit, practical adequacy, report-level sufficiency, reconstructive sufficiency, and identity slide into one another.

The three companion papers isolate different slices of that same pressure. Paper I asks when endpoint agreement is sufficient on a finite window and when a path residual remains. Paper II extracts a general diagnostic framework for separating within-level tension from cross-level residual structure under declared reporting conditions. Paper III shows that the same issue can appear without path accumulation: inside a fixed endpoint slice, histories sharing the same endpoint may still exhibit predictive spread. The shared object is therefore not a loose analogy, but the same cross-level structural pattern appearing through different local carriers.

1.2 Why Endpoint Agreement Is Often Over-Read

Endpoint agreement is often given more weight than the declared setting supports. Once two histories, trajectories, or coarse states share the same endpoint representation, it is tempting to treat that coincidence as settling the comparison. But several stronger claims are being smuggled in: that the endpoint is sufficient for the task, that no reportable residual remains under the chosen discrepancy, that admissible reconstructions do not separate the candidates in a task-relevant way, and that no further structural carrier is needed. None of these follows from endpoint coincidence alone.

What endpoint agreement licenses depends on the locked viewpoint. If the protocol reads only endpoint-level content and the admissible class excludes all task-relevant distinctions inside endpoint fibers, endpoint sufficiency may hold. If task-relevant distinctions remain visible under the same protocol, endpoint agreement does not settle the judgment. The issue is therefore not whether coarse descriptions are legitimate, but whether their legitimacy has been declared rather than presumed.

1.3 From Local Technical Issues to a General Structural Problem

The framework did not begin as an attempt to standardize terminology. It began with repeated cases in which a compressed or endpoint-level description was operationally useful yet struc-

turally incomplete for the intended judgment. In one setting the mismatch appeared between endpoint comparison and pathwise distinguishability. In another it appeared as a cross-level residual left by an apparently natural passage between levels. In a third it appeared as within-endpoint predictive variation that survived coarse representation.

Taken together, these cases point to one structure: a judgment made at one descriptive level is extended across levels without an explicit license, and a residual that remains active under the task is hidden by the convenience of the coarse report. What was needed was therefore not another isolated criterion, but a framework that keeps viewpoint, admissibility, discrepancy, reconstruction, and reporting resolution explicit.

1.4 Scope of This Overview

This article is not a literature survey and does not re-prove the companion papers. Its task is narrower and sharper: to reconstruct the common problem pressure behind them, state the diagnostic discipline they share, and clarify what is already stable across the program at framework level.

Concretely, the overview does four things. It identifies the recurring structural limitation that motivated the framework. It clarifies the role of declared viewpoints, admissibility clauses, reconstruction classes, and resolution-aware reporting. It positions the three papers relative to one another — Paper I as the strict finite-window realization, Paper II as the general framework articulation, and Paper III as the fixed-slice fiber realization. It also clarifies how distinct local realizations belong to one framework without collapsing their local carriers into default identities.

The article should therefore be read not only as a companion overview, but as a clarification of the common structure already visible across distinct diagnostic slices. It does not replace the companion papers. It states more plainly what they jointly establish at the level of cross-level diagnostic criteria, and it does not divide the program into domains that are already unified and domains that only later become part of the same structure.

Paper	Local carrier	What is established	Non-identifications by default
I	endpoint-path residual / geometric proxy	Endpoint sufficiency is a theorem-level question under a locked viewpoint on a finite window.	Not identified by default with ψ , framework-level Δ , or fixed-slice fiber residuals.
II	ψ / Δ	A general cross-level diagnostic framework with declared packages, explicit residual tracking, and separated diagnostic roles.	Not a replacement for every slice-local carrier or for every local theorem proved in Papers I and III.
III	fiber gap / reconstruction residual	Fixed-slice endpoint insufficiency can persist inside predictive fibers even after the endpoint map is declared.	Not identified by default with path accumulation, $\Delta_{\text{acc}}(\gamma)$, or Paper I endpoint-path chain-rule objects.

Table 1: Crosswalk of the three companion papers as distinct but connected diagnostic slices.

Notation and Scope

In this overview, Δ and ψ are the primary framework-level quantities. Informally, ψ records a declared within-level structural tension under a fixed perspective, whereas Δ records a cross-level residual left by a declared passage between levels. They are related, but not interchangeable, and they should not be collapsed into a single undifferentiated notion of complexity or error. At framework level, ψ and Δ organize the diagnostic roles of within-level tension and cross-level residuality; they do not erase the slice-local status of endpoint-path residuals, geometric proxies, fixed-slice fiber gaps, or reconstruction residuals. No default identification among these carriers is asserted here.

Any auxiliary parameter such as γ is used only in local subcontexts. It may index an evolution, refinement depth, or scale parameter inside a paper-specific module, but it is secondary to the framework-level role of Δ and ψ . Unless explicitly stated otherwise, near-zero statements for Δ and ψ are read relative to a declared scale, reporting convention, or admissibility condition. Thus $\Delta \approx 0$ and $\psi \approx 0$ are report-level statements under locked clauses, not free-standing ontological identifications.

2 The Original Problem Pressure

2.1 When a Coarse Description Appears Adequate but Remains Structurally Incomplete

The framework emerged from a recurring asymmetry between operational adequacy and structural completeness. In many settings a coarse description works: it supports a prediction, stabilizes a comparison, or summarizes a task effectively. The problem begins when that local success is upgraded into the stronger conclusion that the coarse object exhausts the structure relevant to the judgment.

That upgrade is often unwarranted. A compressed report may succeed because the task is narrow, because the interface reads only certain observables, or because the reporting threshold is too coarse to resolve what remains underneath. None of these implies that the suppressed structure has disappeared. It implies only that the current regime does not register it. A coarse object can therefore be legitimate as a report while remaining structurally incomplete.

2.2 Hidden Assumptions Behind Cross-Level Comparison

Many cross-level comparisons rely on assumptions that remain active in the reasoning but absent from the statement. The main ones are visibility, admissibility, reconstruction, and negligibility. The report presupposes what counts as visible structure, which objects and transports belong to the comparison class, whether any reverse passage is licensed, and under what threshold a residual may be treated as negligible.

These assumptions do not merely accompany cross-level comparison; they partly constitute it. Once they are made explicit, some comparisons remain valid and others do not.

2.3 Why Sufficiency, Comparability, and Reconstruction Cannot Remain Implicit

Sufficiency, comparability, and reconstruction are not neutral background notions. They determine what a cross-level judgment means. If they remain implicit, the statement itself becomes unstable.

To call a coarse object sufficient is to say that it is sufficient for a declared task, under a declared discrepancy, relative to a declared admissible class, and at a declared reporting threshold. To call two reports comparable is to say that they belong to the same licensed comparison regime. To treat a coarse object as reconstructively informative is to say that a reconstruction discipline has been declared. Once these clauses are omitted, a useful report is easily mistaken for complete closure.

2.4 From Domain-Specific Puzzles to a Shared Diagnostic Pattern

The same pattern kept reappearing in settings that initially looked unrelated. In one case the issue was endpoint agreement versus path-sensitive distinguishability. In another it was a cross-level remainder left by a passage between levels. In another it was predictive spread inside endpoint fibers. The local objects differed, but the diagnostic grammar did not.

A coarse or declared object was treated as sufficient before the conditions of that sufficiency had been stated, and a residual active under the task was suppressed because it did not appear in the most convenient summary. The framework emerged to make that remainder explicit, localize it, and prevent cross-level claims from inheriting undeclared assumptions.

3 Semantic Commitments of the Framework

3.1 Viewpoint Locking as a Structural Condition of Reportability

A central commitment of the framework is that viewpoints are not external commentary added after the formal statement. They are part of the statement. A cross-level comparison is always made under a declared interface, discrepancy rule, admissible class, reconstruction discipline, and reporting threshold. Change those clauses and the semantic status of the comparison changes with them.

This requirement serves two purposes. It blocks silent shifts in meaning, and it makes comparability reproducible. Without a locked viewpoint, one may prove a statement under one interface, read it under another, and treat the result as unchanged. With a locked viewpoint, one can say whether two reports are comparable, merely analogous, or not licensed for comparison at all.

Viewpoint dependence here is not arbitrariness. A viewpoint is one of the conditions under which structure becomes reportable. The stronger claim is therefore not just that reports should mention their viewpoints, but that cross-level structure does not become determinately reportable independently of a declared regime of access, admissibility, comparison, and scale. In that sense, viewpoint locking belongs to the framework's conditions of reportability rather than to an optional layer of expository caution.

3.2 Why the Coupling Role of V Has Philosophical Weight

The coupling role of V matters not only methodologically but philosophically. It clarifies that the framework is not reducing objectivity to perspective, nor treating viewpoint as a merely subjective add-on. The point is narrower and stronger: finite reports are produced from within reality rather than from outside it. Structure therefore does not enter a report in a viewpoint-free way. It becomes reportable only under a declared regime of access, admissibility, comparison, reconstruction, and scale.

This is why boundaries first appear as boundaries of description without being merely subjective artifacts. When a report reaches a point at which a single coarse passage no longer remains licensed, the obstruction is not exhausted by saying that language failed. The report has encountered a structural segmentation in the object under the conditions by which the object was being accessed. In that sense, the coupling role of V makes explicit a deeper claim already implicit in the three companion papers: reportable structure arises through the meeting of objective organization and situated access, not through a view from nowhere.

3.3 Reportability Versus Ontological Identity

The framework also separates reportability from ontological identity. In practice, one often moves too quickly from a report such as “the residual is negligible” or “the endpoint is sufficient” to the stronger claim that the compared objects are simply the same. The framework blocks that move unless it is independently licensed.

A reportable near-zero residual means only that, under a declared package and resolution, the residual lies below the reporting threshold. It does not mean that no structurally relevant difference exists under every package or at every finer scale. Likewise, endpoint sufficiency under a declared package means sufficiency for that task under those clauses; it does not mean that the endpoint exhausts all relevant structure in every regime.

The same caution applies to cross-level equalities. Such equalities often function as shorthand for report-level equivalence under declared conditions. The framework allows that shorthand only when its semantics are locked; it does not promote it by default to unrestricted identity.

3.4 Resolution-Relative Approximation and the Status of “Approximately Zero”

Expressions such as $\Delta \approx 0$ and $\psi \approx 0$ are treated as controlled report-level statements. Their meaning depends on a declared threshold, admissible class, and discrepancy rule. “Approximately zero” is therefore not a metaphysical weakening of equality, but a reporting convention under explicit conditions.

This matters because cross-level diagnostics operate at finite resolution. A residual may be negligible for one declared purpose and non-negligible for another. A fiber may be effectively constant under one discrepancy and clearly non-constant under another. The framework therefore insists on two constraints: every near-zero statement inherits its meaning from a declared reporting convention, and the authority of that report ends where the convention ends.

3.5 Undefinedness, Incommensurability, and Package-Locked Statements

Not every verbally formulable comparison is licensed as a meaningful report. Sometimes the relevant package lacks admissibility clauses, domain restrictions, transport rules, or reconstruction permissions. In such cases the correct output is not an improvised scalar surrogate but an explicit non-licensing state.

This is one of the framework’s main protections against false precision. If two reports are made under different packages without a declared bridge, their comparison may be incommensurable rather than merely difficult. If a discrepancy is not licensed, the corresponding residual should not be smuggled in informally. If reconstruction classes are not declared, claims about recoverability remain incomplete. Package-locked statements may be fully meaningful under one regime while remaining undefined outside it.

Taken together, these commitments define the framework’s diagnostic discipline: viewpoints are locked, reportability is separated from identity, approximation is resolution-relative, and undefinedness is accepted when licensing conditions are absent.

4 Paper I: Endpoint-Path Diagnostics on Finite Windows

4.1 The Endpoint Sufficiency Problem

Paper I isolates a precise question: on a fixed finite window, when is endpoint-only comparison sufficient for the declared discrimination task, and when does that compression leave behind a residual that remains structurally active? The point is not that paths always matter, nor that endpoints are usually enough. The point is that, under a fixed viewpoint, endpoint sufficiency is itself a theorem-level claim.

The diagnostic is formulated under an explicitly locked viewpoint

$$V = (O, \text{protocol}, R, S, \text{threshold rule}, M, \Pi_{\text{rec}}, D, \text{Agg}).$$

Once this specification is fixed, distinguishability on the window becomes a V -conditioned question. Endpoint matching then means not merely that two coarse endpoint reports look similar, but that the endpoint retains all information still distinguishable under the declared interface and protocol. Paper I turns that claim into a precise criterion.

This matters because endpoint agreement is routinely used in RG/EFT practice as if it already settled finite-window consistency. Paper I shows that such a reading is justified only under a stronger condition: the compared scale-chain processes must become indistinguishable once conditioned on the endpoint pair. If that condition fails, endpoint agreement is only a partial report. Paper I is therefore the strict finite-window anchor of the broader question later generalized by the $\Delta\text{-}\psi$ framework.

4.2 Why Path Accumulation Cannot Be Collapsed by Default

The central structural result of Paper I is that endpoint discrepancy and full path discrepancy do not coincide by default. Under the strict bridge one obtains

$$\Delta_{\text{path}} = \Delta_{\text{pair}} + \Gamma_{\text{path}}, \quad \Gamma_{\text{path}} \geq 0.$$

Here Δ_{pair} is the endpoint-pair marginal divergence and Γ_{path} is the residual contributed by the path-conditioned part of the comparison. The key point is semantic as much as algebraic: Γ_{path} is not a nuisance term, but the precise remainder left after compressing a scale chain to its endpoints under the locked viewpoint.

Accordingly, $\Gamma_{\text{path}} = 0$ if and only if the endpoints are sufficient statistics for distinguishing the compared scale-chain processes under that viewpoint. If $\Gamma_{\text{path}} > 0$, intermediate-scale structure remains distinguishable and cannot be compressed into endpoints at the declared resolution.

Paper I also distinguishes route-dependent endpoint quantities. The implementation-level endpoint divergence Δ_I and the strict bridge term Δ_{pair} may coincide only under an explicitly declared implementation. They are not interchangeable by default. This distinction already anticipates a core rule of the later framework: once the viewpoint is locked, one must keep separate what is visible at endpoints, what accumulates along the path, and what remains as residual after compression.

4.3 The Meaning of Endpoint-Path Residuals

The residual Γ_{path} should be read as a structural diagnostic of compressibility on a finite window. It measures historically distinguishable content discarded when a scale chain is reduced to its endpoints under the fixed specification. A positive residual does not announce an unrestricted failure of RG/EFT. It says something narrower and sharper: endpoint-only compression is insufficient for the declared comparison task on the declared window.

Paper I also constructs a computable geometric proxy. Route I defines the endpoint divergence Δ_I , while Route II induces an information metric and information speed Ψ , whose integral yields the path length L_{II} . Under the additional geometric validity conditions one may define

$$\Gamma_{\text{geo}} := L_{II} - d_{\text{end}},$$

where d_{end} is the induced endpoint distance. Within the declared validity regime, $\Gamma_{\text{geo}} = 0$ means that the trajectory segment is a minimizing geodesic between the endpoints.

The paper is equally explicit that Γ_{geo} is not the same object as Γ_{path} . The former is a computable proxy under Route-II validity conditions; the latter is the strict chain-rule residual in the endpoint-path bridge. Any relation between them requires additional assumptions. This non-identification preserves the hierarchy between strict diagnostic content and operational proxy.

4.4 What This Paper Establishes, and What It Does Not Attempt to Establish

Paper I establishes four things. First, it makes the viewpoint part of the mathematics rather than an expository afterthought. Second, it proves the strict endpoint-path bridge and the associated nonnegative residual with an exact sufficiency criterion. Third, it gives a computable geometric proxy with invariant meaning and a sharp zero criterion within its declared validity regime. Fourth, it provides reproducible calibrations in standard finite QED and EFT windows.

Its limits are equally important. It does not yet provide a general framework for cross-level residuals outside the endpoint-path setting. It does not separate, at framework level, within-level tension from cross-level residual mismatch. It does not treat fixed-slice fiber ambiguity. And it does not claim unconditional equivalence between the strict residual and the geometric proxy.

For the present overview, Paper I should therefore be read as the finite-window strict anchor of the program. It establishes the first indispensable claim: endpoint sufficiency must be proved rather than presumed, and the question can be made exact through a viewpoint-conditioned endpoint-path bridge with a sharp residual criterion.

5 Paper II: Generalizing the Problem into a Multi-Level Diagnostic Framework

5.1 From Specific Path Residuals to General Cross-Level Residual Structure

Paper II begins from a limitation already visible in Paper I. The endpoint-path bridge gives a sharp finite-window criterion, but it is tied to one local architecture: path-sensitive comparison under a fixed interface, together with a strict decomposition and, where available, a geometric proxy. What remained open was how to restate the same pressure outside that specific setting.

Paper II addresses exactly that problem. Its aim is not to replace the endpoint-path diagnostic, but to extract the more general framework that Paper I exhibited only locally. The decisive question is still whether a declared report at one level remains licensed when transported across levels, and what residual remains once the relevant clauses are made explicit.

For that reason, Paper II shifts from one specific residual to a general diagnostic pair. Under a declared package $\mathcal{P}$, it tracks both a within-level structural quantity and a cross-level structural quantity. The earlier endpoint-path residual becomes one important instance of a broader phenomenon: a cross-level remainder left by declared transport, compression, identification, or report. Paper II therefore marks the transition from a local strict theorem to a general framework for cross-level residual structure.

5.2 Intra-Level Tension and Cross-Level Residuals

The central conceptual contribution of Paper II is the explicit separation between ψ and Δ . They are related, but they are not interchangeable and should not be collapsed into a single

notion of complexity or error.

Under a declared viewpoint V and package $\mathcal{P}$, ψ records within-level structural tension. It measures nontrivial organization internal to a declared level, relative to what has been fixed as visible, admissible, and reportable. By contrast, Δ records a cross-level residual. It appears when one attempts to relate, transport, compress, or identify structures across levels and some non-collapsible remainder is left under the declared package. In that sense, Δ is not a generic error term; it is the framework’s explicit carrier of non-closure under declared cross-level passage.

This distinction allows the framework to say cleanly what Paper I only suggested locally. A system may exhibit low reportable ψ at a chosen level while still exhibiting non-negligible Δ under cross-level transport. Conversely, substantial within-level tension need not produce a large cross-level residual. Paper II therefore treats ψ and Δ as a coupled diagnostic pair while refusing to identify them by default.

5.3 Declared Packages, Admissibility, and Reproducible Reporting

A further advance of Paper II is that it places package declaration at the center of the diagnostic procedure. Every report of ψ , Δ , comparability, sufficiency, or approximation inherits its meaning from a declared package $\mathcal{P}$. The package is not optional annotation; it determines what counts as a legitimate comparison in the first place.

Depending on the setting, a package may specify an observable interface, admissible object class, transport rule, reconstruction discipline, discrepancy rule, scale convention, and reporting threshold. Once these clauses are locked, diagnostic statements become reproducible. Reports made under the same package are comparable as statements of the same type. Reports made under different packages require an explicitly declared bridge.

This package-centered discipline also clarifies admissibility. A quantity is not reportable merely because one can write it down; it is reportable only if the package licenses the comparison and the discrepancy can be interpreted under the declared interface. The same is true of reconstruction. Expressions such as $\psi \approx 0$ or $\Delta \approx 0$ are therefore not casual summaries, but package-conditioned outputs under declared thresholds and admissibility rules.

5.4 Static and Dynamic Readings of the Framework

Paper II also clarifies that the framework admits both static and dynamic readings. In the static reading, one fixes a package and compares levels, descriptions, or reports without introducing an explicit evolution parameter. The main outputs are then package-conditioned values or statuses such as reportable ψ , reportable Δ , admissibility, undefinedness, and locked comparability.

The dynamic reading adds a declared evolution, refinement, or continuation parameter, denoted locally by γ . One then tracks how the reportable status of ψ and Δ changes along that declared parameterization. This dynamic reading is derivative of the static one: it is a parameterized extension of package-locked reporting, not a replacement for it.

That claim is not merely promissory. The companion framework already contains a minimal evolution interface in which declared intra-level readout can be accumulated along an explic-

itly declared parameter, while remaining distinct from fixed-slice inter-level diagnostics. The dynamic side is therefore slim but genuine.

This layered reading also clarifies the relation to the other papers. Paper I is a windowed or path-sensitive realization under a fixed viewpoint. Paper III is a largely static or fixed-slice realization in which the main issue is predictive spread inside endpoint fibers rather than path accumulation. Paper II supplies the framework in which both can be described without forcing them into one local formal mold.

5.5 What Became General Here That Was Only Implicit Before

Paper II makes several previously local lessons explicit. First, residual structure becomes a first-class diagnostic object: whenever a cross-level statement leaves a non-collapsible remainder under the declared package, that remainder must be tracked rather than absorbed into informal interpretation.

Second, viewpoint locking becomes full package-based determination. What appeared in Paper I as a carefully specified comparison protocol is recognized here as a general requirement of cross-level reporting. Third, the paper separates within-level structure from cross-level residual structure in a way the first paper did not yet need to do explicitly. Without that separation, one cannot distinguish cases in which local structure remains active without producing a large cross-level remainder from cases in which a cross-level mismatch persists despite a seemingly tame within-level profile.

Finally, Paper II enlarges the framework’s output space. The result of a diagnostic procedure need not be a single scalar. Depending on the declared package, the correct output may be a near-zero report, a non-negligible residual, an admissibility failure, a package-locked statement, or an undefined comparison. For the present overview, Paper II is therefore the point at which the program becomes explicitly framework-theoretic.

6 Paper III: Fixed-Slice Fiber Diagnostics and Hierarchical Attention

6.1 Why Fixed-Slice Analysis Is Not the Same as Path Analysis

Paper III studies a different diagnostic slice from Paper I. Its central question is not whether a scale-chain path leaves an irreducible residual after endpoint compression on a finite window. It asks instead: once an endpoint representation $\text{End} : H \rightarrow E$ has already been fixed, how much predictive ambiguity can remain among histories that share the same endpoint value? The analysis therefore works within endpoint fibers rather than along a path in scale or hierarchy.

This distinction is explicit. The diagnostic is fixed-slice: comparisons are made among histories satisfying $\text{End}(h_1) = \text{End}(h_2)$ at a single endpoint value, not by decomposing information along a multi-stage path. The relevant objects are fiber diameters, global gaps, and fixed-slice reconstruction residuals, not endpoint-path chain-rule quantities. Appendix A therefore states a non-identification boundary: the bare symbols Δ_I and Γ_{path} are reserved for endpoint-path

diagnostics and are not reused in the fiber setting.

Without that boundary, Paper III could be misread as merely rewriting Paper I in machine-learning language. But the two questions are different. Paper I asks whether path-sensitive distinguishability collapses to endpoints under a fixed viewpoint. Paper III asks whether a fixed endpoint map is thin enough to support effectively single-valued prediction on its fibers.

6.2 Fibers, Endpoint Collisions, and Within-Endpoint Predictive Spread

The central objects of Paper III are endpoint fibers and the predictive spread they support. Once an endpoint map $\text{End} : H \rightarrow E$ is fixed, each endpoint value $e \in E$ induces a fiber

$$\mathcal{F}(e) := \text{End}^{-1}(e),$$

namely the histories indistinguishable at endpoint level. The paper then compares the protocol-visible predictive distributions $p(\cdot \mid h)$ among histories in the same fiber.

The point is simple. Endpoint agreement does not imply predictive agreement. Two histories may collide at the endpoint level while still inducing substantially different predictive distributions under the declared protocol. The quantity $\text{Gap}_{TV}(\text{End}; C)$ measures the worst-case such non-constancy on the admissible class C . The thresholded statement

$$\text{Gap}_{TV}(\text{End}; C) < \varepsilon$$

is therefore the package-conditioned criterion for calling the endpoint ε -sufficient.

The paper also introduces the fixed-slice minimax reconstruction residual

$$\text{Res}_{TV}(\text{End}; C, U_E) := \inf_{u \in U_E} \sup_{h \in C} TV(p(\cdot \mid h), u(\text{End}(h))),$$

which asks how well one can reconstruct a representative predictive distribution from the endpoint alone under the declared admissible family. The global fiber gap and this minimax residual bound one another, so within-fiber predictive spread and endpoint-based reconstruction loss are two views of the same fixed-slice obstruction.

6.3 Why Hierarchical Refinement Can Help

Against that background, the slogan “hierarchical attention works” receives a precise structural reading. Hierarchical or multi-scale mechanisms are modeled as endpoint families $\{\text{End}_K\}_{K \geq 0}$. Each added scale, summary, memory tier, or long-range anchor augments a baseline endpoint representation. When the augmentation is genuinely refining on C , fibers become thinner because histories previously grouped together are now split by the refined endpoint.

This yields the paper’s monotonicity principle. If End_{K+1} refines End_K on C , then the set of history pairs sharing an endpoint can only shrink, so the global gap cannot increase. Hierarchical refinement is therefore not a vague appeal to “more context.” It is a structural operation that reduces within-fiber predictive non-constancy when the added coordinates are aligned with the operative dependency.

The paper is equally clear that not every refinement helps. In aligned refinement, the added coordinates track the relevant dependency and the fiber gap drops, possibly below the declared threshold ε . In misaligned refinement, critical coordinates remain uncarrried and the gap can plateau. This is why the paper includes capacity-style lower bounds and uses KL-based bridges only when KL is explicitly licensed under the declared package.

6.4 What This Application Reveals About the Broader Framework

Paper III is not merely an application note. It reveals a distinct slice of the same program. The cross-level problem diagnosed earlier does not always appear first as path accumulation or endpoint-path decomposition. It can also appear in fixed-slice form: endpoint agreement has already been declared, yet protocol-visible predictive ambiguity remains active inside endpoint fibers.

Within the Ψ - Δ language, Paper III aligns most naturally with fixed-slice inter-level diagnostics such as $\Delta_{I \rightarrow G}$ rather than with $\Delta_{\text{acc}}(\gamma)$ or endpoint-path residuals. Its local statement $\text{Gap}_{TV}(\text{End}; C) \approx_\varepsilon 0$ means only that the endpoint is reportable as ε -sufficient under the locked package. It does not claim unrestricted identity.

The paper also identifies a concrete trigger for structural extension. If endpoint fibers still support non-negligible predictive variation under the declared query family, then endpoint-level description is insufficient for single-valued audit or predictive closure. This is exactly the fixed-slice pressure that the broader framework must be able to represent.

7 How the Three Papers Form One Research Program

7.1 A Common Logical Core

The three companion papers use different local constructions, but they are driven by the same logical core. In each case, a statement made at a coarse or declared level is liable to be over-read unless the conditions of its validity are explicitly locked. The issue is not merely that coarse descriptions lose information. The issue is whether the declared coarse object is sufficient for the declared task, what residual remains if it is not, and under what conditions the result is reportable at all.

The grammar is stable across all three papers. A diagnostic begins with a declared interface: what is visible, what is admissible, what discrepancy is licensed, what reconstruction class is allowed, and what threshold governs reporting. A coarse object is then introduced. The question is whether that object closes the intended judgment under those conditions. If it does, sufficiency may be reported at the declared resolution. If it does not, the remainder must be tracked as residual, reported as non-negligible, or marked as undefined when the relevant clauses are missing.

7.2 Different Diagnostic Slices: Path, Cross-Level, and Fiber

The three papers operate on different diagnostic slices. Paper I studies the path slice: the coarse object is an endpoint pair extracted from a scale-chain process, and the central issue is whether endpoint comparison is sufficient on a finite window. The residual appears through the endpoint-path bridge, so the diagnostic is path-sensitive and window-conditioned.

Paper II studies the cross-level framework slice. It abstracts away from any one local realization and asks how residual structure should be described once one no longer assumes that the decisive issue is specifically path accumulation. This is where the distinction between ψ and Δ becomes indispensable and where the output space broadens beyond a single scalar summary.

Paper III studies the fiber slice. The coarse object is an endpoint representation already fixed in advance, and the residual appears as within-endpoint ambiguity rather than path accumulation. The relevant objects are endpoint fibers, fiber gaps, and fixed-slice reconstruction residuals. What unifies the slices is not a shared formula, but a shared architecture: coarse sufficiency is conditional, residuals are explicit, and viewpoint locking is indispensable.

7.3 Shared Semantic Discipline Across the Papers

The unity of the program lies in a stable diagnostic discipline. First, all three papers lock the relevant viewpoint/package. None treats the comparison protocol as an inessential wrapper. Second, all three separate reportability from unrestricted identity. Whether the local form is endpoint sufficiency, near-zero Δ , or ε -sufficient endpoint representation, the report remains tied to declared conditions.

Third, all three localize non-closure rather than letting it disappear into rhetoric. In Paper I it is localized in the endpoint-path residual. In Paper II it appears as Δ or, when appropriate, as explicit non-licensing. In Paper III it is localized on endpoint fibers through the global gap and fixed-slice reconstruction residual. Fourth, all three maintain domain discipline: local claims are issued only under their declared validity conditions.

7.4 Why the Framework Had to Be Split Across Multiple Articles

The framework had to be split across multiple articles because a single paper could not have carried all three slices without losing precision. Each slice requires its own local objects, domain assumptions, and form of proof. The endpoint-path paper needs strict path-process objects and a separate computable proxy. The fiber paper needs endpoint families, predictive interfaces, fiber gaps, reconstruction classes, and refinement monotonicity. Compressing both into one article would have obscured their boundary conditions and encouraged false identifications among deliberately distinct objects.

The split reflects expository separation rather than theoretical sequencing. Each article fixes one local carrier under its own boundary conditions so that the shared structure can be stated without collapsing distinct objects into one premature formal mold. Paper I isolates the strict finite-window endpoint-path slice. Paper II isolates the framework-level diagnostic pair and package-locked reporting conditions. Paper III isolates the fixed-slice fiber slice. The split is

therefore not merely editorial; it preserves local precision while leaving the shared structure unchanged.

8 Framework-Level Content and Cross-Domain Realization

8.1 Framework-Level Components Established Across the Papers

Several components of the framework are already established across the companion papers. First, viewpoint/package declaration belongs to the diagnostic content itself. Second, coarse sufficiency is a theorem-level or package-level claim rather than a default inference from coarse agreement. Third, residual structure must be localized explicitly rather than absorbed into informal interpretation. Fourth, the distinction between within-level structure and cross-level residual structure is stabilized through the separation between ψ and Δ . Fifth, reportability itself is sharpened: near-zero statements are package- and resolution-relative, while undefinedness, non-licensing, and package-locked comparability are legitimate outputs of diagnosis.

Taken together, these elements define the framework-level content clarified here: declared comparability, testable sufficiency, explicit residual localization, separation of ψ from Δ , and resolution-aware reporting.

8.2 Further Local Realizations Under the Same Diagnostic Discipline

What follows are not domains waiting to be unified, but further local realizations of a structure already shared across the program.

Such realizations may include stronger declared relations among slice-local residual carriers. Paper I distinguishes a strict endpoint-path residual from its geometric proxy. Paper III introduces fiber gaps and reconstruction residuals not identified with path-sensitive objects. Paper II provides the framework in which such objects can coexist without collapsing them into a default identity. The open issue in each case is not whether the structure belongs to the same framework, but which bridges, carriers, and computable realizations are declared in that setting.

They may also include separately declared package-to-package transport regimes and richer dynamic elaborations beyond the minimal interface already established. These directions refine the local realization of the shared structure. They are not presupposed by the framework-level claims summarized in this overview, but neither are they external to the same framework.

8.3 One Structure Across Logic, Physics, Computation, and Learning

The same cross-level structural condition appears in logic, physics, computation, and learning. These are not domains waiting to be unified, but distinct local realizations of one structure. What changes from case to case is the local carrier of non-closure, the declared package, the admissible reconstruction discipline, the reporting resolution, and whether the problem is settled at the level of structural diagnosis or pushed further into explicit quantitative realization.

In logic, the structural condition already appears in compact form: opposing elements + system

self-feedback + an attempted continuous passage from micro-level inferential factors to macro-level closure imply structural break. Once that configuration is present, diagnosis does not depend on a viewpoint-free identity claim. Further quantification remains possible, but only under a declared proof, model, or query interface. In physics, the same structure appears when endpoint or coarse agreement is read as closure too quickly even though pathwise or cross-level residuals remain active under the declared effective description. In computation, the same structure appears when a compressed or protocol-visible representation is treated as sufficient before admissible reconstruction, task visibility, and discrepancy clauses have been fixed. In learning, the same structure appears when endpoint families are treated as if they exhausted predictive content even though within-fiber spread remains active under the declared task.

The difference among these domains is therefore not whether they belong to one framework, but how the shared structure is instantiated locally, how finely it is resolved, and how far the local analysis has been pushed from structural diagnosis into explicit quantitative realization.

8.4 Criteria for Further Local Realizations

A valid further local realization must preserve the diagnostic discipline already established. It must declare its viewpoint/package clearly enough that visibility, admissibility, discrepancy, reconstruction, and reporting thresholds are not left implicit. It must localize residual structure explicitly rather than replacing it with rhetoric or an unlicensed scalar summary.

It must also preserve boundary discipline: if a new object is claimed to relate to Δ , ψ , an endpoint-path residual, or a fixed-slice reconstruction residual, the bridge must be declared rather than presumed. Cross-package transport is best treated as a structural question rather than as a default comparison procedure. It becomes well-posed only when the packages in question can be jointly situated within a declared common regime endowed with unified rules of comparability. Absent such a regime, cross-package comparison remains undefined in the present sense. Any transport law should therefore be formulated as a separately declared higher-order structure, not inferred from the mere coexistence of two packages. Sufficiency and approximation claims must remain package- and resolution-relative unless a stronger theorem has actually been proved. Failure states must be reported as carefully as success states, including non-negligible residuals, undefinedness, and non-licensing.

9 Methodological Scope and Boundary Discipline

9.1 Why Boundary Discipline Belongs to the Framework

This overview does not turn every local result into a premature universal theorem. That is not caution for its own sake. The framework leaves some bridges undeclared, keeps slice-local carriers distinct, and refuses default identifications unless the bridge has been stated. Those boundary conditions belong to the diagnostic rule itself.

For the same reason, package declaration, viewpoint locking, and explicit non-licensing states are not expository decoration. In the companion settings, the deepest failures arose when scope was left undeclared: an equality read as identity, a near-zero report read as collapse, or a slice-

local diagnostic read as if it already transported framework-wide. Precision therefore requires explicit boundary marking because the framework forbids semantic overreach at the level of judgment, not because the underlying structure is only partially shared.

9.2 Limits of the Present Overview

The present article is not a full intellectual history of the framework, nor does it try to write, in one place, every local carrier relation, transport law, reconstruction grammar, or quantitative realization for every domain and every resolution. Some quantities remain module-specific, some bridges remain intentionally undeclared, and some realizations are better introduced in their own local setting. The aim here is to state the common problem pressure, identify the diagnostic commitments stable across the companion papers, and clarify what framework-level content has already been established.

It is also not an exhaustive literature review. Nor does it claim that the structure applies only where the current article has already supplied a detailed local formalization. The structure is already shared across domains; what varies is whether a given case is presently being reported at the level of structural diagnosis, at the level of a declared local diagnostic family, or at the level of a further computable realization.

10 Conclusion

10.1 From Separate Diagnostics to a Unified Framework

The three companion papers should not be read as isolated results connected only by thematic resemblance. They form one research program because they address the same cross-level structural problem through different diagnostic slices. These slices do not form a development ladder; they are distinct local realizations of one framework. In each case, a coarse or declared object is liable to be over-read unless viewpoint, admissibility, discrepancy, reconstruction, and reporting threshold have been locked. What varies across the papers is the local carrier of non-closure; what remains stable is the diagnostic rule by which sufficiency, residual structure, and non-licensing are determined.

Paper I provides a strict finite-window realization: endpoint sufficiency on a finite window is a theorem-level question, and the answer is expressed through a locked endpoint-path bridge with a sharp residual criterion. Paper II extracts the general framework criterion space by separating within-level tension from cross-level residual structure and by making package declaration part of every report. Paper III provides a fixed-slice realization in which the same discipline remains necessary when the residual is localized not on accumulated paths but on endpoint fibers.

10.2 What This Overview Adds Beyond the Three Papers

This article adds more than synthesis. It states explicitly what is already stable at framework level across the three papers.

First, declared viewpoints and packages are conditions of determinate cross-level reporting, not

merely annotations. Second, Δ is not a generic error term but the framework’s explicit carrier of cross-level non-closure. Third, the separation between ψ and Δ distinguishes within-level tension from cross-level residual mismatch and thereby enlarges the legitimate output space of diagnosis beyond a single scalar summary. Fourth, the framework already supports both static and declared dynamic readings through a minimal evolution interface.

The overview also makes explicit a philosophical point that remained only implicit across the three papers: the declared viewpoint/package is not external metadata, but the coupling layer under which structure becomes reportable. This does not reduce objectivity to perspective; it states instead that finite reporting always proceeds from within a situated regime of access. For that reason, viewpoint declaration is not a comment on an already complete result; it is part of the condition under which any cross-level result becomes determinately reportable. The overview should therefore be read as a clarification of the framework-level content the program has already established.

10.3 Framework-Level Content and Further Development

At framework level, the program already fixes declared comparability, conditional sufficiency, explicit residual localization, separation of ψ from Δ , resolution-aware reporting, explicit boundary marking when a comparison is not licensed, and a minimal static-to-dynamic interface. Those commitments are sufficient to organize work across physics, learning, logic, and computation because they define the conditions under which coarse sufficiency, residual activity, approximation, and non-licensing are reported.

Further work under the same diagnostic discipline does not bring previously unrelated domains into the framework. It refines the local realization of the same framework across different objects, resolutions, and declared comparison regimes. Such work may develop stronger declared bridges among local residual carriers, higher-order transport regimes when jointly licensed common systems are available, richer dynamic elaborations, and new domain-specific carriers where needed. These are refinements of shared structure rather than additions of a different kind of problem.

The value of the framework lies not only in the discipline it enforces, but in the criterion space and output structure it already makes available for stating cross-level sufficiency, non-closure, and declared boundary states with precision. In that sense, the framework does more than warn against over-reading coarse agreement. It specifies when coarse agreement is sufficient, when it is not, what carrier remains when it is not, and when the comparison itself is not licensed. It enlarges the space of legitimate mathematical reporting by allowing declared residual carriers, package-conditioned approximation, and explicit non-licensing states to appear as stabilized outputs. It also sharpens a deeper lesson: boundary states first appear as boundaries of description because objective structure becomes reportable only through declared coupling conditions, not through a view from nowhere.

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Data Availability

No new datasets were generated or analyzed in this study.

Conflict of Interest

The author declares no conflict of interest.

References

- [1] Zihe Zhao, *Relative Entropy Endpoint-Path Gaps as a Diagnostic of RG Coarse-Graining Failure*, Zenodo, 2026.
DOI: [10.5281/zenodo.18414030](https://doi.org/10.5281/zenodo.18414030).
- [2] Zihe Zhao, *The Psi-Delta Framework*, Zenodo, 2026.
DOI: [10.5281/zenodo.18456597](https://doi.org/10.5281/zenodo.18456597).
- [3] Zihe Zhao, *Why Hierarchical Attention Works*, Zenodo, 2026.
DOI: [10.5281/zenodo.18810251](https://doi.org/10.5281/zenodo.18810251).